

Kathleen Anderson and Thomas Martinetz

Hidden statistical dependencies exist in almost every dataset. They can encode useful biases, but may also impede the learning process, like an object frequently, but not always, appearing against a specific background or a measurement systematically skewed by the equipment used to generate it. In some cases, these dependencies can even be harmful and lead to unfair or discriminatory outcomes.

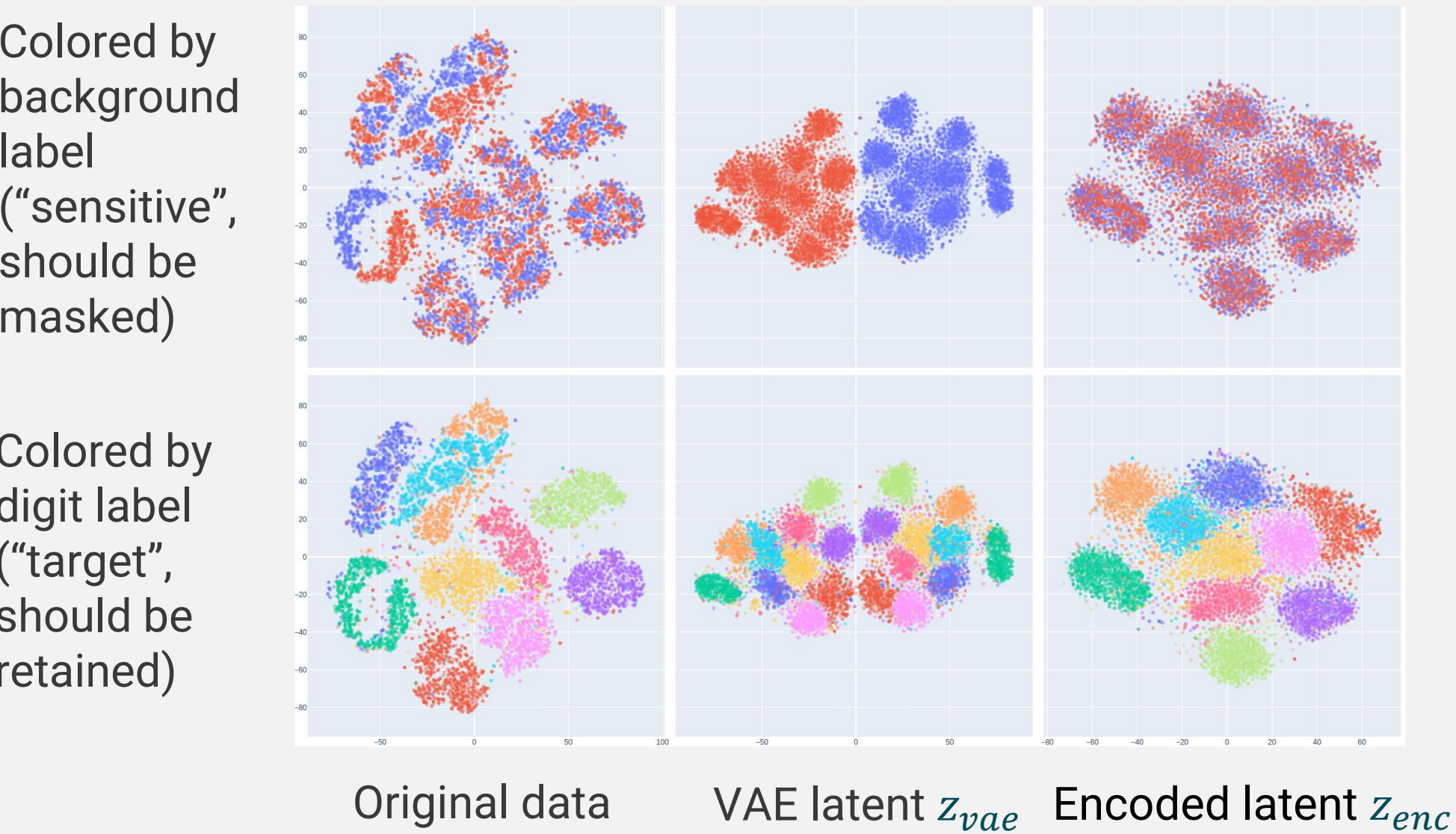
We propose an encoder-based solution that learns a representation independent of a sensitive variable but otherwise preserving essential data characteristics. Eliminating unwanted biases from the dataset before any further model training is performed can prevent a model from internalizing false assumptions from the start.

Results

Accuracy of a classifier trained to predict either a sensitive or a target label, and the difference between the two accuracies.

		Liu 2022	Akash 2021	Ours
MNIST with background shapes	sensitive ↓	51.6	52.7	51.8
	target ↑	95.2	96.9	96.8
	target – sensitive ↓	43.6	44.2	45
FFHQ	sensitive ↓	62.6	65.1	58.2
	target ↑	59.6	59.0	74.8
	target – sensitive ↓	3	6.1	16.6
CheXpert	sensitive ↓	63.8	62.9	66.1
	target ↑	69.9	70.0	74.0
	target – sensitive ↓	6.1	7.1	7.9

T-SNE embedding for the MNIST dataset with shapes added as “sensitive” information to the image background.

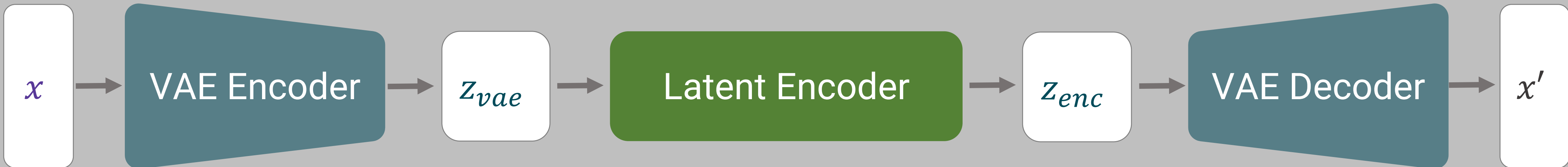


Setting

X: Input samples, continuous and unstructured
S: discrete or discretizable sensitive label (one for each input sample)
Z: learned representation

Goal: minimize the mutual information between sensitive variable and representation, preserve as much about the input as possible.

$$\min_z (I(Z; S) - \beta I(Z; X))$$



Step 1: VAE Disentangling

Standard Variational Autoencoder loss:

$$\mathbb{E}_z \underbrace{\log p(x|z)}_{\text{reconstruction}} - \underbrace{\text{KL}(q(z|x)|p(z))}_{\text{regularization}}$$

The regularization loss pushes the latent distribution to be close to a prior $p(z)$.

Instead of the usual standard normal distribution, we set $p(z)$ to $\mathcal{N}(\mu, I)$, $\mu = [s, 0, \dots, 0]$. The target mean μ is set to zero for all latent dimensions except one. That one dimension is set to the value of the sensitive label.

The encoder is explicitly trained to include information about s into a single latent dimension. Masking this dimension masks most of the sensitive data.

Step 2: Point-Distance Density

The VAE encoder and decoder are frozen; a smaller encoder is placed between them. To minimize the dependency between Z and S further, this encoder is trained to minimize a combination of a reconstruction loss and:

$$I(Z; S) \propto \text{KL}(p(z|s_z); p(z))$$

derivation in the paper

Intuition:
if Z and S are independent, the likelihood of every representation $z \in Z$ is the same when estimated in
(i) the subset of representations whose sensitive label s_z is the same as the sensitive label of z and
(ii) the set of all representations Z .

KL divergence can be minimized with a pointwise estimator based on Kozachenko and Leonenko (1987):

Intuition:
Two distributions are equivalent if for every point z , the distance ε to all neighboring points is the same when estimated in either $p(z)$ or $p(z|s_z)$

$$\text{KL}(p(z|s_z); p(z)) \approx \sum_{z \in Z} p(z|s_z) \log \frac{\varepsilon_{p(z)}(z)}{\varepsilon_{p(z|s_z)}(z)}$$

derivation in the paper